10. $\int_1^2 \frac{8\,dx}{x^2 - 2x + 2}$

$$\int_1^2 \frac{8}{x^2 - 2x + 2}\,dx$$

$$= 8\int_1^2 \frac{1}{(x-1)^2 + 1}\,dx$$

Let x - 1 $= u \implies dx = du$

when x $= 1$, u $= 0$ when x $= 2$, u $= 1$

The integral takes the form

$$8\int_0^1 \frac{1}{u^2+1}\,du$$

$$= 8\tan^{-1} u\Big|_0^1$$

$$= 2\pi$$

$$\approx 6.2832$$

17. $\int \frac{\ln y}{y + 4y\ln^2 y}\,dy$

$$\int \frac{\ln(y)}{y + 4y\ln^2(y)}\,dy =$$

$$\int \frac{\ln(y)}{y(1 + 4\ln^2(y))}\,dy = (\text{☆})$$

u-substitution $u = 1 + 4\ln^2(y) \implies du = 4\frac{2\ln(y)}{y}dy \implies \frac{du}{8} = \frac{\ln(y)}{y}dy$

$$(\text{☆}) = \int \frac{du}{8u} =$$

$$\frac{1}{8}\int \frac{du}{u} =$$

$$\frac{1}{8}\ln|u| + C =$$

$$\boxed{\frac{1}{8}\ln|1 + 4\ln^2(y)| + C}$$

$$\int xe^{3x}\,dx$$

$$= \frac{1}{3}xe^{3x} - \frac{1}{3}\int e^{3x}\,dx \quad \text{using integration by parts}$$

$$= \frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x} + C$$

$$= \frac{1}{9}e^{3x}(3x - 1) + C$$

$$\int \sin^2 x \cos^4 x\,dx.$$

Solution This is an example of Case 3.

$$\int \sin^2 x \cos^4 x\,dx = \int \left(\frac{1 - \cos 2x}{2}\right)\left(\frac{1 + \cos 2x}{2}\right)^2 dx \qquad m \text{ and } n \text{ both even}$$

$$= \frac{1}{8}\int (1 - \cos 2x)(1 + 2\cos 2x + \cos^2 2x)\,dx$$

$$= \frac{1}{8}\int (1 + \cos 2x - \cos^2 2x - \cos^3 2x)\,dx$$

$$= \frac{1}{8}\left[x + \frac{1}{2}\sin 2x - \int (\cos^2 2x + \cos^3 2x)\,dx\right]$$

For the term involving $\cos^2 2x$, we use

$$\int \cos^2 2x\,dx = \frac{1}{2}\int (1 + \cos 4x)\,dx$$

$$= \frac{1}{2}\left(x + \frac{1}{4}\sin 4x\right). \qquad \text{Omitting the constant of integration until the final result}$$

For the $\cos^3 2x$ term, we have

$$\int \cos^3 2x\,dx = \int (1 - \sin^2 2x)\cos 2x\,dx \qquad u = \sin 2x,\ du = 2\cos 2x\,dx$$

$$= \frac{1}{2}\int (1 - u^2)\,du = \frac{1}{2}\left(\sin 2x - \frac{1}{3}\sin^3 2x\right). \qquad \text{Again omitting } C$$

Combining everything and simplifying, we get

$$\int \sin^2 x \cos^4 x\,dx = \frac{1}{16}\left(x - \frac{1}{4}\sin 4x + \frac{1}{3}\sin^3 2x\right) + C.$$ ∎

$$\int \tan^4 x \sec^4 x \, dx.$$

Solution

$$\begin{aligned}
\int (\tan^4 x)(\sec^4 x)\, dx &= \int (\tan^4 x)(1 + \tan^2 x)(\sec^2 x)\, dx && \sec^2 x = 1 + \tan^2 x \\
&= \int (\tan^4 x + \tan^6 x)(\sec^2 x)\, dx \\
&= \int (\tan^4 x)(\sec^2 x)\, dx + \int (\tan^6 x)(\sec^2 x)\, dx \\
&= \int u^4\, du + \int u^6\, du = \frac{u^5}{5} + \frac{u^7}{7} + C && u = \tan x,\ du = \sec^2 x\, dx \\
&= \frac{\tan^5 x}{5} + \frac{\tan^7 x}{7} + C
\end{aligned}$$

- $$\int_{5\pi/6}^{\pi} \frac{\cos^4 x}{\sqrt{1 - \sin x}}\, dx$$

Remaining

$$I = \int \frac{\sqrt{x-2}}{\sqrt{x-1}}\, dx$$

Remaining

$$\frac{-2x+4}{(x^2+1)(x-1)^2} = \frac{Ax+B}{x^2+1} + \frac{C}{x-1} + \frac{D}{(x-1)^2}. \tag{2}$$

Clearing the equation of fractions gives

$$\begin{aligned}
-2x+4 &= (Ax+B)(x-1)^2 + C(x-1)(x^2+1) + D(x^2+1) \\
&= (A+C)x^3 + (-2A+B-C+D)x^2 \\
&\quad + (A-2B+C)x + (B-C+D).
\end{aligned}$$

Equating coefficients of like terms gives

$$\begin{aligned}
&\text{Coefficients of } x^3: & 0 &= A + C \\
&\text{Coefficients of } x^2: & 0 &= -2A + B - C + D \\
&\text{Coefficients of } x^1: & -2 &= A - 2B + C \\
&\text{Coefficients of } x^0: & 4 &= B - C + D
\end{aligned}$$

We solve these equations simultaneously to find the values of A, B, C, and D:

$$\begin{aligned}
&-4 = -2A, \qquad A = 2 && \text{Subtract fourth equation from second.} \\
&C = -A = -2 && \text{From the first equation} \\
&B = (A + C + 2)/2 = 1 && \text{From the third equation and } C = -A \\
&D = 4 - B + C = 1. && \text{From the fourth equation.}
\end{aligned}$$

We substitute these values into Equation (2), obtaining

$$\frac{-2x+4}{(x^2+1)(x-1)^2} = \frac{2x+1}{x^2+1} - \frac{2}{x-1} + \frac{1}{(x-1)^2}.$$

Finally, using the expansion above we can integrate:

$$\begin{aligned}
\int \frac{-2x+4}{(x^2+1)(x-1)^2}\,dx &= \int \left(\frac{2x+1}{x^2+1} - \frac{2}{x-1} + \frac{1}{(x-1)^2} \right) dx \\
&= \int \left(\frac{2x}{x^2+1} + \frac{1}{x^2+1} - \frac{2}{x-1} + \frac{1}{(x-1)^2} \right) dx \\
&= \ln(x^2+1) + \tan^{-1} x - 2\ln|x-1| - \frac{1}{x-1} + C. \quad \blacksquare
\end{aligned}$$

Initial Value Problems

Solve the initial value problems in Exercises 49–52 for y as a function of x.

49. $x\dfrac{dy}{dx} = \sqrt{x^2 - 4}, \quad x \geq 2, \quad y(2) = 0$

$$x\frac{dy}{dx} = \sqrt{x^2-4}$$

$$\implies \frac{dy}{dx} = \frac{\sqrt{x^2-4}}{x}$$

$$\implies dy = \frac{\sqrt{x^2-4}}{x}\,dx$$

Integrating both sides

$$y = \int \frac{\sqrt{x^2-4}}{x}\,dx$$

Let $\mathbf{x} = 2\sec u \implies dx = 2\sec u\tan u\,du$

$$\therefore\ y = \int \frac{\sqrt{4\sec^2 u - 4}}{2\sec u}(2\sec u\tan u)\,du$$

$$= 2\int \sqrt{(\sec^2 u - 1)}(\tan u)\,du$$

$$= 2\int \sqrt{(\tan^2 u)}(\tan u)\,du$$

$$= 2\int \tan^2 u\,du$$

$$= 2\int (\sec^2 u - 1)\,du$$

$$= 2\tan u - 2u + C$$

$$= 2\left(\frac{x^2-4}{2}\right) - 2\sec^{-1}\frac{x}{2} + C$$

$$= \sqrt{x^2-4} - 2\sec^{-1}\frac{x}{2} + C$$

when $\mathbf{x} = 2, \mathbf{y} = 0$

$$\implies 0 = 0 - 2(0) + C \implies C = 0$$

$$\implies y = \sqrt{x^2-4} - 2\sec^{-1}\frac{x}{2}$$

Initial Value Problems

Solve the initial value problems in Exercises 49–52 for y as a function of x.

52. $(x^2 + 1)^2 \dfrac{dy}{dx} = \sqrt{x^2 + 1}, \quad y(0) = 1$

$$(x^2+1)^2\frac{dy}{dx} = \sqrt{x^2+1} \implies \frac{dy}{dx} = \frac{\sqrt{x^2+1}}{(x^2+1)^2} \implies dy = \frac{\sqrt{x^2+1}}{(x^2+1)^2}\,dx$$

integrating both sides

$$y = \int \frac{\sqrt{x^2+1}}{(x^2+1)^2}\,dx$$

$$= \int \frac{1}{(x^2+1)^{\frac{3}{2}}}\,dx$$

Let $x = \tan u \implies dx = \sec^2 u du$

$$\therefore\ y = \int \frac{1}{(\tan^2 u+1)^{\frac{3}{2}}}(\sec^2 u)\,du$$

$$= \int \frac{1}{(\sec^2 u)^{\frac{3}{2}}}(\sec^2 u)\,du$$

$$= \int \cos u\,du = \sin u + C = \frac{x}{\sqrt{x^2+1}} + C$$

when $\mathrm{x} = 0, \mathrm{y} = 1$

$$\implies 1 = 0 + C \implies C = 1$$

$$\therefore\ y = \frac{x}{\sqrt{x^2+1}} + 1$$